

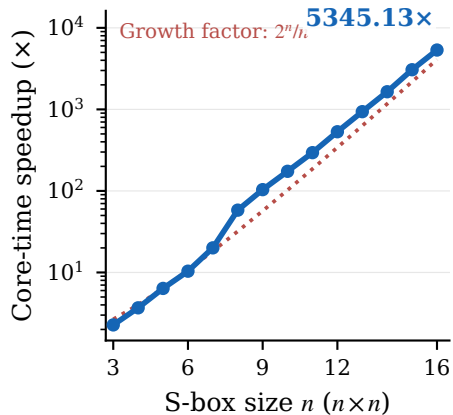
# Algorithmic speedup over the scalar reference

—●— Static / random swaps

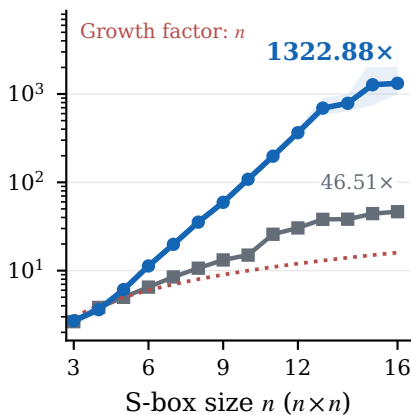
—■— High-update swaps

... Complexity growth factor

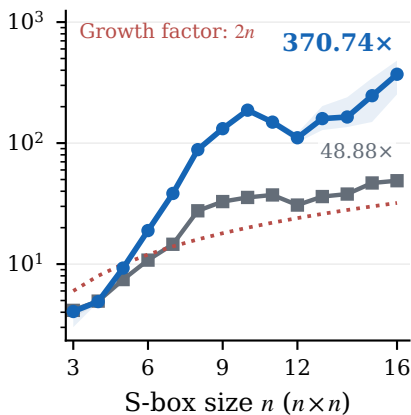
(a) Minimum degree



(b) Maximum degree update



(c) Spectrum update



10 inputs  $\times$  5 repeats. Scalar reference comparison; distinct from PEIGEN / bitwise results.